

A uniform seven-eighths zero-free input in an explicit Chen sieve

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Abstract

We give an explicit reduction from a uniform zero-free half-plane for Dirichlet L -functions to positive Chen representations. If every such function is zero-free in $\Re s > 7/8$, apart from the principal pole at 1, every even $N \geq e^{51046}$ satisfies $D_{1,2}(N) > 9 \cdot 10^{-7} U_N N / (\log N)^2$. The argument uses three Chen sieve terms, exact weights at the smallest primes, and finite Rosser weights at medium primes. We derive an effective prime-character estimate from the zero-free input, account for small primitive conductors even when induced to large moduli, and prove a fixed-cutoff prime–integer rectangle estimate. Complete finite prime enumerations and outward-rounded interval integration certify the starting inequalities; analytic monotonicity propagates them to the whole half-line. The external zero-free assertion is kept separate from the finite certificates. No GRH distribution estimate is used.

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1 Result and proof plan

Let $\Omega(n)$ count prime factors with multiplicity, and for even N put

$$D_{1,2}(N) = \#\{p < N : p \text{ prime, } N - p \geq 2, \Omega(N - p) \leq 2\}.$$

Write $L = \log N$ and

$$C_\infty = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right), \quad C_N = C_\infty \prod_{\substack{p>2 \\ p|N}} \frac{p-1}{p-2}, \quad U_N = 2e^{-\gamma}C_N, \quad \Theta_N = \frac{U_N N}{L^2}.$$

The analytic input to be used is the following precise assertion:

$$L(s, \chi) \neq 0 \quad (\Re s > 7/8), \quad \text{except for the principal pole at } s = 1. \quad (\text{Z})$$

It is uniform over all Dirichlet characters, moduli, and heights.

Theorem 1.1. *Under (Z), every even integer $N \geq e^{51046}$ satisfies*

$$D_{1,2}(N) > 9 \cdot 10^{-7} \Theta_N > 0.$$

Theorem 1.1 of the external manuscript [1] asserts exactly (Z); adopting that theorem removes the hypothesis of Theorem 1.1. Its proof is not certified by our interval programs. Our citation fixes a source revision rather than treating a downloaded formalization entry point as a locally checked proof of its dependencies.

There are two different roles for the zero-free input. It controls a bounded range of conductors inside the averaged ordinary AP estimate, and it controls the low-conductor part of the switched rectangles. High conductors are handled by published mean estimates and the multiplicative large sieve. Summing a bound of order $x^{7/8}$ over $\sqrt{x}$ moduli would not give the distribution estimate we need.

The combinatorial reduction has one positive sieve term and two negative ones. Exact/finite presieving makes their distribution support inexpensive; it replaces a large primorial with a small exact product and a bounded medium-prime weight. The last negative term is a three-prime sequence. Its smallest prime is separated into short intervals so that the other two primes give a coefficient sequence bounded by one in each rectangle. Section 7 gives the complete normalized ledger.

All parameters used in the theorem are fixed rational numbers:

$$\begin{aligned} L_0 = 51046, \quad \alpha = \frac{2158047}{500000000}, \quad t = \frac{1}{2} - \alpha, \quad D = N^t, \quad z = N^{1/8}, \quad y = N^{1/3}, \\ T = 5000000, \quad w_0 = 5, \quad u = 11/2, \quad D_0 = T^u, \quad \kappa = 11/169, \quad Q_* = 15, \\ \varepsilon = 1/41219, \quad \eta_+ = 1/3231, \quad \eta_- = 1/3242, \quad \delta = 10^{-6}, \quad \ell = \log(1 + \delta), \quad v = 10^{-8}. \end{aligned} \quad (1)$$

Here D is the large-prime Rosser level, D_0 the defining medium level, and κD_0 its actual support bound. Neither is the conductor cutoff D_{R0} introduced in Section 5. The certificate proves $C_\infty > 0.66$ by a finite product and an integer tail, hence throughout we use $U_N > U_* := 1.32e^{-\gamma}$.

2 An effective consequence of the zero-free input

We first supply the character estimate needed later. Its deliberately generous constants avoid unspecified implied constants. For a character χ modulo q , set $\psi(x, \chi) = \sum_{n \leq x} \Lambda(n) \chi(n)$ and $\pi(x, \chi) = \sum_{p \leq x} \chi(p)$.

Lemma 2.1. Assume (Z). For $q \geq 1$, $x \geq 2$, and every character modulo q , with $a = 61/64$ and $\delta_\chi = 1$ for the principal character and 0 otherwise,

$$|\psi(x, \chi) - \delta_\chi x| \leq 10^6 x^a \log(50qx), \quad |\pi(x, \chi) - \delta_\chi \text{li}(x)| \leq 10^7 x^a \log^2(50qx).$$

Both bounds hold at left limits as well.

Proof. For nonprincipal χ , its periodic partial sums have absolute value at most q . Partial summation gives $|L(s, \chi)| \leq q|s|/\sigma$ for $\sigma = \Re s > 0$. For a principal character use the holomorphic normalization $K(s) = (s-1)L(s, \chi)/(s+1)$, taking its removable value $\varphi(q)/(2q) > 0$ at 1; otherwise put $K = L$. The identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{r\} r^{-s-1} dr$$

and the omitted Euler factors give the same growth bound for this K . In detail, $|s|/|s+1| \leq 1$ for $\Re s > 0$, and their product has absolute value at most $2^{\omega(q)} \leq q$.

On the closed disk centered at $2 + i\tau$ of radius $71/64$, $\Re s \geq 57/64 > 7/8$, so K is holomorphic and nonzero and $|K(s)| \leq 10q(|\tau| + 5)$. Choose $g = \log K$ using the Euler logarithm at the center. Its absolute value there is less than 4: the Euler logarithm has absolute value less than 1, and the additional principal logarithm has absolute value at most $\log 3 + \pi/2$. Put $M = \log(10q(|\tau| + 5)) \geq \log 50$. Borel–Carathéodory on the inner radius $141/128$ gives

$$|g| \leq 282(M + 4) + 4 < 600M.$$

Every point $\sigma + i\tau$ with $29/32 \leq \sigma \leq 2$ is at distance at most $140/128$ from the center. Cauchy’s formula on its radius $1/128$ gives $|g'| < 76800M$. At $\sigma = 29/32$, the extra principal logarithmic derivatives have absolute sum at most $32/3 + 32/61 < 12$. Thus

$$|L'/L(29/32 + i\tau, \chi)| < 80000 \log(10q(|\tau| + 5)).$$

On horizontal segments of the strip the same bound holds for $|\tau| \geq 1$, with the pole correction at most 2.

For $\psi_1(x, \chi) = \sum_{n \leq x} (x-n)\Lambda(n)\chi(n)$, the absolutely convergent smoothed Perron identity is

$$\psi_1(x, \chi) = \frac{1}{2\pi i} \int_{(2)} -\frac{L'}{L}(s, \chi) \frac{x^{s+1}}{s(s+1)} ds.$$

Move the contour to $29/32$. There are no zeros in the strip; only the principal pole contributes, with residue $x^2/2$. The horizontal integrals tend to zero, since for fixed x their integrands are bounded by a constant times $\log(q(|\tau| + 5))/\tau^2$. The denominator on the new line is at least $.81 + \tau^2$ in absolute value. With $B = \log(50q)$,

$$\int_0^\infty \frac{\log(10q(\tau + 5))}{.81 + \tau^2} d\tau \leq (100/81 + 1)(B + \log 2) + 1 < 3B.$$

Here use $\log(1 + \tau) \leq \log 2$ on $[0, 1]$ and $\log(1 + \tau) \leq \log 2 + \log \tau$ on $[1, \infty)$. Since $240000/\pi < 10^5$,

$$|\psi_1(x, \chi) - \delta_\chi x^2/2| < 10^5 x^{61/32} \log(50q).$$

Take $h = x^{61/64} \leq x$. The forward difference quotient differs from $\psi(x, \chi)$ by at most $(h+1)\log(x+h)$. The two smoothed errors divided by h total less than $5 \cdot 10^5 x^a \log(50q)$, since $1 + 2^{61/32} < 5$. The principal main term adds $h/2$. As $h \geq 1$ and $\log(x+h) \leq \log(50qx)$, these terms imply the first bound.

Removing prime powers costs at most $\sqrt{x}(\log x)^2/\log 2 \leq 2x^a \log^2(50qx)$. Thus the corresponding θ error has constant $10^6 + 2$. Partial summation from 2 bounds the endpoint and integral by $x^a \log^2(50qx)$ times

$$\frac{1}{\log 2} + \frac{1}{a \log^2 2} < 4.$$

The principal integration constant $2/\log 2 - \text{li}(2)$ has absolute value less than 4, absorbed by 10^7 . No primitivity has been assumed: the periodic sum and omitted Euler factors use the actual modulus q . Left limits follow by continuity of the error bounds. $\square$

Put $r_0 = 3/64$ and define, for $x \geq 2$,

$$\tau(x) = 10^7 x^{-r_0} \left[(1 + r_0^{-1}) \log^2(50x) + \frac{2 \log(50x)}{r_0^2} + \frac{2}{r_0^3} \right]. \quad (2)$$

Lemma 2.2. *For $L \geq L_0$, the following inputs hold under (Z):*

$$\begin{aligned} \left| \sum_{p < x} \frac{1}{p} - \log \log x - M \right| &\leq \tau(x), \quad \tau(x) < L^{-2} \quad (x \geq z), \\ N/L < |\mathcal{A}| &< (1 + 2/L)N/L, \quad \pi(x) < (1 + 4/L)x/\log x \quad (x \geq y), \\ \frac{U_N}{\log x}(1 - v) &< V_N(x) < \frac{U_N}{\log x}(1 + v) \quad (x = z, y). \end{aligned}$$

Here $\mathcal{A} = \{N - p : p < N, p \nmid N\}$, M is the Meissel–Mertens constant, $\mathcal{P}_N$ is the set of odd primes not dividing N , and $V_N(x) = \prod_{p < x, p \in \mathcal{P}_N} (1 - 1/(p-1))$. If W is a nonnegative monotone kernel on $[b, c]$ with $z \leq b < c$, then

$$\left| \sum_{b \leq p < c} \frac{W(p)}{p} - \int_b^c \frac{W(x)}{x \log x} dx \right| \leq 2L^{-2} \max\{W(b), W(c)\}. \quad (3)$$

Proof. Partial summation of Lemma 2.1 with $q = 1$ gives the reciprocal-prime error $E(x) = e(x)/x - \int_x^\infty e(t)t^{-2} dt$, where $e(x) = \pi(x) - \text{li}(x)$. The integral converges and its bound is exactly (2); the constant is M by taking limits. The Stieltjes identity gives (3), because the sum of endpoint contributions and total variation of W is $2 \max W$.

Elementary envelopes, starting at $\log x = 20$, are

$$\frac{x}{\log x} \left(1 + \frac{1}{\log x} \right) < \text{li}(x) < \frac{x}{\log x} \left(1 + \frac{1}{\log x} + \frac{3}{\log^2 x} \right).$$

The starting comparison is an interval evaluation of $\text{Ei}(20)$; the two derivatives minus $\text{li}'(x)$ are respectively $-2/\log^3 x$ and $(\log x - 9)/\log^4 x$. Set $e_0(b) = 10^7 e^{-3b/64} (b + \log 50)^2$. The three mass conclusions follow respectively from

$$L^2 e_0(L) + \frac{L^3 e^{-L}}{\log 2} < 1, \quad L^2 e_0(L) + 3/L < 1, \quad L(L/3) e_0(L/3) + 27/L < 1.$$

The first pays for every prime divisor of N . All three conditions are certified at L_0 ; their positive terms decrease thereafter.

The constant matches the Euler product through

$$M = \gamma + \sum_p (\log(1 - 1/p) + 1/p).$$

For completeness, compare $\log \zeta(1 + h)$ as $h \downarrow 0$. Partial summation of $\sum_{p \leq x} 1/p = \log \log x + M + o(1)$ gives $\sum_p p^{-1-h} = \log(1/h) - \gamma + M + o(1)$, using $\int_0^\infty e^{-u} \log u du = -\gamma$. The

absolutely convergent terms with prime-power exponent at least two in $\log \zeta(1+h)$ tend to $-\sum_p (\log(1-1/p) + 1/p)$, whereas $\log \zeta(1+h) = \log(1/h) + o(1)$. Comparing the constants proves the identity. The logarithmic error of the Mertens product is consequently at most $\tau(x) + 1/(x-1)$, using the convergent expansion of $-\log(1-1/p)$. The product factorization yields

$$\frac{V_N(x) \log x}{U_N} = e^\gamma \log x \prod_{p < x} (1 - 1/p) \frac{\prod_{2 < p < x} (1 - (p-1)^{-2})}{C_\infty} \prod_{\substack{p|N \\ p \geq x}} \frac{p-2}{p-1}.$$

The middle factor has logarithm between 0 and $1/(x-2)$. The last loses at most $2L/x$ in the present range. Both product errors are bounded by

$$e^{\tau(z)+4/z} - 1 + 4L/z < v.$$

The certificate also checks $\tau(z) < L^{-2}$. These conditions continue to hold: their positive terms are bounded by polynomials in L times $e^{-3L/512}$ or $e^{-L/8}$, with negative derivatives already at L_0 . This uses no RH Mertens estimate. $\square$

3 Uniform products and finite presieving

This section establishes weights whose coefficients are bounded by one and whose support fits the later average distribution estimates. Only the primes through w_0 are presieved exactly; it is not necessary to multiply every prime through T into an exact primorial.

3.1 A product condition at every intermediate endpoint

For $b > 2$ put

$$\mathcal{D}(b) = \log b \prod_{2 < p < b} (1 - 1/(p-1)), \quad \mathcal{D}_\infty = 2e^{-\gamma} C_\infty.$$

The external unconditional inputs, for $b \geq 10^4$ and in the endpoint convention of limits, are [2, Lemma 16 (finite upper bound) and Lemma 17 (lower bound)] and [4, Lemma 3.2]:

$$E(b) := \sum_{p < b} 1/p - \log \log b - M \geq -2.964 \cdot 10^{-6} / \log b,$$

$$E(b) \leq 2/(\sqrt{b} \log b) \quad (b \leq 10^{12}), \quad E(b) \leq 6.836 \cdot 10^{-6} / \log b \quad (b > 10^{12}).$$

Let $b_A = \max\{2/(\sqrt{A} \log A), 6.836 \cdot 10^{-6} / \log(10^{12})\}$ and $k_A = b_A + 2.964 \cdot 10^{-6} / \log A + 1/(A-1) + .54/A$. For $A \geq 10^4$ even, these inputs imply

$$\prod_{\substack{b \leq p < c \\ p > A}} (1 - 1/(p-1))^{-1} \leq e^{k_A} \frac{\log c}{\log b} \quad (1 < b < c). \quad (4)$$

Indeed $\log(\mathcal{D}(b)/\mathcal{D}_\infty) = -E(b) + \sum_{p \geq b} \rho(p)$, where $\rho(p) = -\log(1-1/(p-1)) - 1/p > 0$ and $\rho(p) \leq 1/[p(p-2)] + .54/p^2$ for $p > A$. The two integer tails sum to at most $1/(A-1) + .54/A$. Use the upper bound at c and the lower bound at b . For $b \leq A < c$ take $b' \downarrow A$; for $c \leq A$ the product is empty.

To sharpen this uniformly, take $S = 10^8$ and $Q = 10^7$, both composite, and the finite lower bound

$$\lambda_Q = \log 2 - \gamma + \sum_{2 < p \leq Q} \log(1 - (p-1)^{-2}) + \log(1 - 1/Q) \leq \log \mathcal{D}_\infty.$$

The inequality follows from the all-integer telescoping tail $\prod_{n>Q}(1 - (n-1)^{-2}) = (Q-1)/Q$. Put $d(b) = \log \mathcal{D}(b)$ and bound the following quantities:

$$A_T \geq \sup_{T \leq b < c \leq S} (d(b) - d(c)), \quad M_T \geq \sup_{T \leq b \leq S} (d(b) - \lambda_Q), \quad B_T = \max\{A_T, M_T + b_S, k_S\}.$$

Finite–finite pairs are covered by A_T ; finite–infinite pairs by $M_T + b_S$, since $d(c) - \log \mathcal{D}_\infty \geq -b_S$ for $c > S$; infinite–infinite pairs by (4) at S . Between primes d increases, and it jumps down just after each prime. Thus every ordered pair is covered by every left-limit peak followed by every right-limit trough and the two composite endpoints. For adjacent intervals their peak/trough/drawdown triples combine as $(\max P_i, \min Q_i, \max\{A_1, A_2, P_1 - Q_2\})$. This is a complete finite algorithm, not endpoint sampling. The certified conclusion is

$$\prod_{\substack{b \leq p < c \\ p > T}} (1 - 1/(p-1))^{-1} < (1 + \varepsilon) \frac{\log c}{\log b} \quad (1 < b < c). \quad (5)$$

Deletion of primes preserves it. The finite certificate enumerates all 5761454 odd primes through S and proves $e^{B_T} - 1 < 0.000024260536 < \varepsilon$.

3.2 Medium weights and first failure

For any deletion of the complete primes $w_0 < p \leq T$, let P_0 be their product and $g(p) = 1/(p-1)$. For $d = p_1 \cdots p_n$, $p_1 > \cdots > p_n$, set $b^\pm(d) = \mu(d)$ precisely when

$$p_1 \cdots p_n p_m^2 < D_0$$

at every odd $m \leq n$ for the upper sign, and every even $m \leq n$ for the lower sign; otherwise set it to zero. The empty weight is 1. Write $B^\pm = \sum_{d|P_0} b^\pm(d)g(d)$ and $V_0 = \prod_{p|P_0} (1 - g(p))$.

Lemma 3.1. *The weights bracket the roughness indicator pointwise. Their masses obey*

$$B^+ = V_0 + \sum_{n \text{ odd}} T_n, \quad B^- = V_0 - \sum_{n \text{ even}} T_n,$$

where T_n sums $g(p_1 \cdots p_n) \prod_{p < p_n, p|P_0} (1 - g(p))$ over prefixes whose first checked failure is at length n . Every supported divisor and every first-failing prefix has product $< D_0$. Supported divisors in fact satisfy $d < \kappa D_0$.

Proof. Expand the full inclusion–exclusion product and partition omitted terms by their first checked failure. For a fixed such prefix, the unrestricted smaller-prime suffix sums to the displayed product. Its sign is positive in $B^+ - V_0$ and in $V_0 - B^-$ because of the relevant parity. The same finite identities hold at $g(p) = 1_{p|m}$ and imply the pointwise brackets; all these first-failure products are nonnegative.

For a supported divisor, if its last prime r is checked, $dr^2 < D_0$. If it is unchecked and follows prime $s > r$, the preceding checked prefix gives $d < D_0 r/s^2$. A single unchecked prime is at most T . For a first failure of length at least three, its preceding checked prefix is two primes shorter, ending at $s > q > r$, and $d = d_{n-2}qr < d_{n-2}s^2 < D_0$. Lengths one and two have product at most $T^2 < D_0$. This proves the product bound for every first failure.

For the sharper support use

$$\kappa = \max \left\{ p_0^{-2}, \max_{w_0 < r < s \leq T} \frac{r}{s^2} \right\},$$

with p_0 the smallest complete medium prime. The checked and unchecked cases above give $d < \kappa D_0$; also $T < \kappa T^u$ since $\kappa \geq T^{-2}$ and $u > 3$. Deletion cannot increase this complete-set maximum. For each s only its preceding complete prime need be checked. Exact integer comparisons over all adjacent pairs give $\kappa = 11/169$, attained at (11, 13). $\square$

3.3 A positive block bound for both masses

This supplies a finite reproducible bound without dropping the preceding Rosser prefix checks. Let $r_1 < r_2 < \dots$ be all medium primes and let K be the largest integer with $r_1 \dots r_K < D_0$. Take $h = 1/200$ and $J = \lceil \log D_0/h \rceil$. Label a prefix by $B = \sum \lfloor \log p/h \rfloor$ and group primes in descending blocks of equal individual label b . For each block set

$$c = \left\lceil \frac{\log D_0 - 3 \log p_{\min}}{h} \right\rceil, \quad f = \max \left\{ 0, \left\lceil \frac{\log D_0 - 3 \log p_{\max}}{h} \right\rceil - K + 1 \right\}.$$

For each sign maintain two nonnegative polynomials (X_0, X_1) , initially $(1, 0)$, for the parity of accepted prefixes. Define A to append label b , flip parity, and at a checked append retain only source labels $B < c$. Truncate output labels to $B < J$. Let $\mathcal{F}$ sum source coefficients $B \geq f$ of the parity checked by one append. With $t_p = g(p)/(1 - g(p)) = 1/(p - 2)$ set

$$D_b(a) = \prod_{p \text{ in block}} (1 + t_p + t_p a), \quad C_b(a) = \frac{D_b(a) - 1}{1 + a}.$$

Both have nonnegative coefficients. Replace X by

$$\sum_{j=0}^K [a^j] D_b(a) A^j X,$$

and add $\mathcal{F} \sum_{j=0}^K [a^j] C_b(a) A^j X$ to the sign's gap total.

Lemma 3.2. *These two totals bound $(B^+ - V_0)/V_0$ and $(V_0 - B^-)/V_0$, uniformly under every deletion of medium primes. Their certified values satisfy*

$$(B^+ - V_0)/V_0 < 0.000309498615 < \eta_+, \quad (V_0 - B^-)/V_0 < 0.000308430014 < \eta_-.$$

Proof. If a prefix has k chosen primes and logarithm v_1 , then $hB \leq v_1 < h(B + k)$. Every true pass satisfies $B < c$. A first failure has $k \leq K - 1$ preceding primes and $v_1 + 3 \log p \geq \log D_0$, which implies $B \geq f$. All actual prefixes have product $< D_0$ by Lemma 3.1, hence degree at most K and label $< J$. The relaxed tests therefore contain every actual pass and failure. Counting an ambiguous label both ways only increases positive bounds.

Before processing a prime multiply the accepted-prefix mass by $\prod_{q>p} (1 - g(q))^{-1}$. One prime then gives

$$X' = ((1 + t_p)I + t_p A)X, \quad G' = G + t_p \mathcal{F}X.$$

The augmented matrices commute within a block since $A, \mathcal{F}$ are common. Their product gives $X' = D_b(A)X$ and $G' = G + \mathcal{F}C_b(A)X$. Equivalently $D_{\text{new}} = (1 + t + ta)D$ and $C_{\text{new}} = (1 + t + ta)C + t$, starting from $D = 1, C = 0$. If e_m are elementary symmetric sums of t_p , then

$$[a^j]D_b = \sum_{m \geq j} \binom{m}{j} e_m, \quad [a^j]C_b = \sum_{m \geq j+1} \binom{m-1}{j} e_m.$$

Terms $j > K$ cannot contain an actual contribution. Artificial paths of excessive total degree may survive and harmlessly enlarge the bound. Deletion decreases every coefficient of D_b, C_b ; retaining the complete blocks and their positive operators proves deletion uniformity.

The finite evaluation generates all 348510 medium primes, takes $K = 22, J = 16968$, and processes all 1810 nonempty blocks. Every logarithmic integer cutoff is resolved by an Arb interval with a unique integer floor or ceiling. Polynomial coefficients use positive Arb addition and multiplication. This evaluates exactly the finite majorant above and proves the stated strict rational comparisons. $\square$

3.4 Composite weights and the large-prime linear sieve

The external large-prime input is the Rosser main-mass estimate in [2, Theorem 6, Proposition 13 and Table 1] with $Q = 1$. Its local density satisfies $0 \leq g(p) < 1$, and its dimension-one product premise holds at every intermediate endpoint by (5). With $s = \log D / \log x$, the masses $G_1^\pm$ satisfy

$$G_1^- \geq V_1(x)l(s), \quad G_1^+ \leq V_1(x)u(s), \quad l(s) = f(s) - 108\varepsilon e^2 h(s), \quad u(s) = F(s) + 106\varepsilon e^2 h(s).$$

The lower estimate needs $D \geq x^2$ and the upper $D \geq x$. The coefficients $\lambda_1^\pm$ are $\mu(d)$ or zero, have support $d < D$, and their divisor sums bracket the corresponding roughness indicator. In the ranges used here,

$$F(s) = 2e^\gamma/s \quad (1 \leq s \leq 3), \quad f(s) = 2e^\gamma \log(s-1)/s \quad (2 \leq s \leq 4),$$

$$F(s) = \frac{2e^\gamma}{s} \left(1 + \int_3^s \frac{\log(x-2)}{x-1} dx \right) \quad (3 \leq s \leq 4), \quad h(s) = \begin{cases} e^{-2} & 1 \leq s \leq 2, \\ e^{-s} & 2 \leq s \leq 3, \\ 3e^{-s}/s & s \geq 3. \end{cases}$$

The table constants (106, 108) are valid since $\varepsilon < 10^{-4}$.

Let Q_N contain the retained odd primes through 5, so $Q_N \mid Q_*$, and let $\mu_E(d) = \mu(d)1_{d|Q_N}$. Define

$$\lambda^+ = \mu_E * b^+ * \lambda_1^+, \quad \lambda^- = \mu_E * (b^+ * \lambda_1^- - (b^+ - b^-) * \lambda_1^+).$$

For pointwise brackets $a^- \leq I_0 \leq a^+$ and $c^- \leq I_1 \leq c^+$, with upper brackets nonnegative,

$$a^- c^+ + a^+ c^- - a^+ c^+ \leq I_0 I_1 \leq a^+ c^+.$$

The lower expression is at most $I_0 I_1 - (a^+ - I_0)(c^+ - I_1)$. Multiplication by exact roughness gives the displayed composite weights. Their prime ranges are disjoint, so factorization is unique. Write $b^\pm(d_0) = \mu(d_0)I^\pm$ and $\lambda_1^\pm(d_1) = \mu(d_1)J^\pm$ with indicators in $\{0, 1\}$. The lower normalized coefficient $I^+ J^- - (I^+ - I^-)J^+$ belongs to $\{-1, 0, 1\}$, by the two cases $J^+ = 0, 1$. Thus all full coefficients have absolute value at most one and support $d < \kappa Q_* D_0 D$.

For any nonnegative finite sequence $\mathcal{C}$ with $|\mathcal{C}_d| = |\mathcal{C}|/\varphi(d) + r_{\mathcal{C}}(d)$, put $R = \sum_{d < \kappa Q_* D_0 D, d|P_N(x)} |r_{\mathcal{C}}(d)|$. The main-mass calculation gives

$$\begin{aligned} S(\mathcal{C}, x) &\geq |\mathcal{C}|V_N(x)\{(1 + \eta_+)l(s) - (\eta_+ + \eta_-)u(s)\} - R, \\ S(\mathcal{C}, x) &\leq |\mathcal{C}|V_N(x)(1 + \eta_+)u(s) + R. \end{aligned} \tag{6}$$

Indeed, with $a_0 = (B^+ - V_0)/V_0$, $b_0 = (V_0 - B^-)/V_0$, the normalized lower mass is at least $(1 + a_0)l - (a_0 + b_0)u$. It decreases in a_0, b_0 because $u \geq l$ and $u > 0$. The positive lower-mass condition is certified for the actual s . Expanding the bounded coefficients against the sequence gives R .

4 Ordinary AP distribution

Put $x_1 = N/L^5$, $w = \log x_1 = L - 5 \log L$, and $H = \sqrt{x_1}/\log^{10} x_1$. The distribution cost is averaged over squarefree moduli, not estimated one modulus at a time by Lemma 2.1. The external ordinary tools in [2, Lemmas 22–24] give

$$\begin{aligned} \omega(d) &< 1.3841 \log d / \log \log d, \\ \sum_{d \leq R} \mu^2(d) &\leq .65R \quad (R \geq 45), \end{aligned}$$

$$\sum_{d \leq R} \mu^2(d)/\varphi(d) \leq 1.1 \log R \quad (R \geq 10^9).$$

The first bound is used for $d \geq 3$. The high-primitive-conductor input is the last displayed estimate in the proof of [2, Lemma 30, source label `small1313`], obtained there from Akbary–Hambrook [3, Theorem 1.2] and partial summation. For $Q = \log^{10} x_1$, $x_1 \leq Y \leq N$, it gives

$$\log Y \sum_{\substack{Q < r \leq H \\ (r, N) = 1}} \frac{\mu^2(r)}{\varphi(r)} \sum_{\chi \bmod r}^* |\psi(Y, \chi)| \leq 27\mathcal{E}(Y), \quad (7)$$

where

$$\mathcal{E}(Y) = \frac{4Y \log^{9/2} Y}{\log^{10}(Y/\log^5 Y)} + \frac{4Y}{\log^{11/2} Y} + \frac{18Y^{11/12}}{\sqrt{\log Y}} + \frac{5}{2} Y^{5/6} \log^{11/2} Y.$$

Its required starting condition is $\log \log x_1 \geq 10.4$ and $Q < H$; both are verified here. The restriction $(r, N) = 1$ is retained exactly as in the source: it holds automatically when $r \mid d$ and $(d, N) = 1$ in the application below. This high-conductor input is unconditional and contains no zero term; the exceptional-modulus restriction on the full BJS lemma is not used to estimate this isolated primitive high-conductor sum. Stars mean primitive characters; in this range none is principal.

Lemma 4.1. *Under (Z), for $L \geq L_0$ and $r(d) = |\mathcal{A}_d| - |\mathcal{A}|/\varphi(d)$,*

$$R_A := \sum_{\substack{d \leq H \\ (d, N) = 1}} \mu^2(d) |r(d)| \leq c_4(N) N/L^3, \quad c_4(N) L^2 < 2.21.$$

The estimate includes all squarefree moduli, not only $d \mid P_N(z)$.

Proof. In character orthogonality write every nonprincipal character modulo d as an induction of a unique primitive character modulo $r \mid d$. Its induction error in $\psi(Y)$ is at most $\omega(d) \log Y \leq \log^2 Y / (2 \log 2)$. The total such error is at most $\sqrt{Y} / (2 \log 2 \log^8 Y)$ since $H \leq \sqrt{Y} / \log^{10} Y$. The principal deletion carries $1/\varphi(d)$, and totals less than $.4 \log^3 Y$.

Exact conductor regrouping of the remaining terms is bounded by

$$A(H) \sum_{r \leq H} \frac{\mu^2(r)}{\varphi(r)} \sum_{\chi \bmod r}^* |\psi(Y, \chi)|, \quad A(H) = \sum_{m \leq H} \mu^2(m)/\varphi(m) \leq .55 \log Y.$$

For $r \leq Q$, Lemma 2.1, the bound of $\varphi(r)$ characters, and $\log(50QY) < 2 \log Y$ give $1.1 \cdot 10^6 Y^{61/64} (\log Y)^{12}$ for the entire low-conductor sum, including its inductions to large d . At w , the certificate checks $1.1 \cdot 10^6 e^{-3w/64} w^{20} < 3.2 \cdot 10^{-8}$. The left side decreases for $w > 1280/3$; thus this cost is at most $3.2 \cdot 10^{-8} Y / \log^8 Y$. The program retains the larger allowance $1.1 \log(\log^{10} Y) 3.2 \cdot 10^{-8} Y / \log^8 Y$. For $r > Q$, use $A(H) \leq .55 \log Y$ and (7); the resulting cost is at most $.55 \cdot 27\mathcal{E}(Y) < 27\mathcal{E}(Y)$. The resulting four positive costs are exactly the zero-term-free ψ bound used in the next formula.

Convert ψ to θ and then π as in [2, Lemma 31, source label `small1533`]. The prime-power bound is $1.02\sqrt{Y} + 3Y^{1/3}$. Brun–Titchmarsh on the initial segment x_1 costs $2.2x_1$. With $E_\theta(Y; d) = \theta(Y; d, N) - \theta(Y)/\varphi(d)$, the exact partial summation identity used is

$$E_\pi(N; d) = E_\pi(x_1; d) + \frac{E_\theta(N; d)}{L} - \frac{E_\theta(x_1; d)}{w} + \int_{x_1}^N \frac{E_\theta(Y; d)}{Y \log^2 Y} dY.$$

The integral is bounded by its elementary antiderivative majorant $N/[L^4(1 - 4/w)]$. Put

$$E_w = \frac{4w^{9/2}}{(w - 5 \log w)^{10}} + 4w^{-11/2} + 18e^{-w/12} w^{-1/2} + \frac{5}{2} e^{-w/6} w^{11/2},$$

$$\begin{aligned}
p_2 &= w^2 \left(1.1 \cdot 10 \log w \frac{3.2 \cdot 10^{-8}}{w^8} + 27E_w + \frac{e^{-w/2}}{2 \log 2 w^8} + .4w^3 e^{-w} \right), \\
p_1 &= p_2 + (.67 + 2e^{-w/6})w^{-8}, \\
p &= p_1 \left(1 + \frac{1}{L^2 w^3} + \frac{1}{(1 - 4/w)L} \right) + 2.2/L^2, \\
c_4(N) &= p + .9e^{w/2-L} \frac{L^4}{w^{10} \log L}.
\end{aligned}$$

All positive p_2 summands decrease as functions of $\log Y \geq w$; therefore their supremum is the displayed endpoint. Finally deleting the primes dividing N costs at most $1.3841L/\log L$ per modulus, and at most $.65H$ moduli occur. This gives the last summand of c_4 , since $.65 \cdot 1.3841 < .9$. This deletion argument uses only squarefreeness and $(d, N) = 1$, so the narrower printed sieve-modulus range in BJS Lemma 43 is unnecessary. The starting inequality $c_4(N)L^2 < 2.21$ is certified; its half-line extension is proved in Section 7.1. $\square$

5 A fixed-cutoff rectangle estimate

The third sieve term has a prime in a short interval and an integer coefficient formed from two larger primes. We need a uniform estimate whose low-conductor input remains valid at its much smaller endpoints. Put $X_3 = N^{1/8}/2$, $x_3 = \log X_3$, and $D_{R0} = x_3^{10}$. For $r \leq \log^{10} X_3$ and $b \geq X_3$, Lemma 2.1 gives

$$\sum_{\chi \bmod r, \chi \neq \chi_0} |\pi(b, \chi)| \leq v_3 \frac{b}{\log^5 b}, \quad v_3 = 4 \cdot 10^7 X_3^{-3/64} x_3^{17}. \quad (8)$$

Indeed $\log(50rb) < 2 \log b$, there are at most r characters, and $e^{-3\xi/64} \xi^{17}$ decreases for $\xi > 1088/3$.

Lemma 5.1. *Let $X = N/w_j$, $Z = w_j$, $Y = \min(y, (1+\delta)w_j)$, where $z \leq w_j < y$. For $|a(n)| \leq 1$, the sum of the absolute nonprincipal rectangle errors over squarefree $d \leq H$ with $d \mid P_N(y)$ is at most*

$$\begin{aligned}
&39v_3XY/\log^3 Y + 108X \log^{13} Y/\log \log Y \\
&+ 26XY \log^2 Y/D_{R0} + 88XY \log^2 Y(X^{-1/2} + Y^{-1/2}) + 106XY/\log^9 Y.
\end{aligned}$$

The error is centered at $\varphi(d)^{-1} \sum_{n < X} \sum_{Z \leq p < Y, (np, d)=1} a(n)$.

Proof. Here $XY \geq N$, $XY \leq (1+\delta)N$, $X > N^{2/3}$ and $Y > z$. Consequently $H < \sqrt{XY}/\log^{10} Y$ and $H \leq Y^4$. Also $10^9 < D_{R0} < z < H$ and $Z > X_3$. Orthogonality and primitive conductor decomposition $d = sr$ give

$$\sum_{sr \leq H} \frac{1}{\varphi(sr)} \sum_{\chi \bmod r}^* \left| \sum_{n < X, (n, s)=1} a(n) \chi(n) \right| \left| \sum_{Z \leq p < Y, p \nmid s} \chi(p) \right|.$$

All moduli remain squarefree; hence $(r, s) = 1$ and $\varphi(rs) = \varphi(r)\varphi(s)$. For $r < D_{R0}$, use (8) at both left limits of Z, Y . Since $b/\log^5 b$ increases in this range, their total is at most $2v_3Y/\log^5 Y + \varphi(r)\omega(s)$. The integer sum is at most X . Two reciprocal-totient sums give

$$2.42v_3XY \log^2 H/\log^5 Y + 1.21 \cdot 1.3841D_{R0}X \log^3 H/\log \log H.$$

In bounding $\omega(s)$ by $1.3841 \log H/\log \log H$, use monotonicity of $\log b/\log \log b$ for $b \geq 16$. For $1 \leq s \leq 15$, $\omega(s) \leq 3$ and the same endpoint bound is greater than 3, so the small values

introduce no exception. Use $H \leq Y^4$, $D_{R0} \leq \log^{10} Y$, and monotonicity of $\log^3 b / \log \log b$ for $b \geq 5$. The first two constants become $2.42 \cdot 16 < 39$ and $1.21 \cdot 1.3841 \cdot 64 < 108$.

For high conductors split into half-open intervals $R_j \leq r < \min(2R_j, H)$, with $R_j = 2^j D_{R0}$. The number of nonempty intervals is at most $\log H / \log 2$ because $D_{R0} \geq 2$, including the last interval, and $\sum_j R_j < 2H$. The multiplicative large-sieve input in the proof of [2, Lemma 32, source label lemma:bf] is

$$\sum_{R \leq r < 2R} \frac{r}{\varphi(r)} \sum_{\chi}^* \left| \sum_{n < X} b_n \chi(n) \right|^2 \leq (X + 12R^2) \sum_{n < X} |b_n|^2.$$

Use it for the integer and prime sequences, whose squared norms are at most X and Y . Cauchy's inequality bounds one interval by

$$XY/R + \sqrt{12XY} + \sqrt{12Y}X + 12R\sqrt{XY}.$$

Sum over s at cost $1.1 \log H$, the first three terms by the number of intervals and the fourth by their geometric sum. With $\log H \leq 4 \log Y$ and $H \leq \sqrt{XY} / \log^{10} Y$, the constants are $1.1 \cdot 16 / \log 2 < 26$, $1.1 \cdot 16 \sqrt{12} / \log 2 < 88$, and $1.1 \cdot 4 \cdot 24 = 105.6 < 106$. This proves every term of the estimate. The final prime interval may be truncated; none of these bounds assumes that it has the full relative width δ . $\square$

6 Three Chen terms and their main masses

Define $P_N(b) = \prod_{p < b, p \in \mathcal{P}_N} p$ and $S(\mathcal{C}, b) = \#\{n \in \mathcal{C} : (n, P_N(b)) = 1\}$. For every positive d , let $\mathcal{A}_d = \{a/d : a \in \mathcal{A}, d \mid a\}$; in particular this defines $\mathcal{A}_q$ for $q \in \mathcal{P}_N$. The pointwise combinatorial input is Nathanson's Theorem 10.2 as restated in [2, Lemma 37, source label lem:>]:

$$D_{1,2}(N) \geq S(\mathcal{A}, z) - \frac{1}{2} \sum_{z \leq q < y, q \nmid N} S(\mathcal{A}_q, z) - \frac{1}{2} S(\mathcal{B}, y) - 2N^{7/8} - N^{1/3} - 1, \quad (9)$$

where $\mathcal{B} = \{N - p_1 p_2 p_3 : z \leq p_1 < y \leq p_2 \leq p_3, p_1 p_2 p_3 < N, (p_1 p_2 p_3, N) = 1\}$. The extra 1 safely excludes $N - p = 1$ under either almost-prime convention. The exponent $7/8$ here comes from a combinatorial truncation, not from (Z). We retain it in full.

For the lower sieve use level D at z , with $s = 8t \in (3, 4)$. For the prime-factor sum use level D/q , giving $s_q = 8(t - \log q/L) \in [k, 3]$, $k = 8(t - 1/3) \in (1, 2)$. For $\mathcal{B}$ use level D at y , giving $3t \in (1, 2)$. The conditions $D \geq z^2$, $D/q \geq z$, $D \geq y$ follow from these ranges. The common distribution support is legitimate because

$$\alpha L - \frac{25}{2} \log L > \log(\kappa Q_*) + u \log T, \quad H > \sqrt{N}/L^{25/2}. \quad (10)$$

The first is certified at L_0 and increases thereafter. Thus $\kappa Q_* D_0 D < H$, and the $\mathcal{A}_q$ support is below H/q .

6.1 The positive term

Equation (6) and Lemma 2.2 give

$$S(\mathcal{A}, z)/\Theta_N \geq M_0 - R_A/\Theta_N, \quad M_0 = 8(1 - v)\{(1 + \eta_+)l(s) - (\eta_+ + \eta_-)u(s)\}.$$

Its bracket is strictly positive, as certified.

6.2 The prime-factor term: a full representative calculation

Fix $q \in [z, y)$ not dividing N . The sequence $\mathcal{A}_q$ has model mass $|\mathcal{A}|/(q-1)$ and level D/q . For $d \mid P_N(z)$,

$$r_q(d) = r(qd) - r(q)/\varphi(d), \quad |\mathcal{A}_q| = |\mathcal{A}|/(q-1) + r(q).$$

Its main mass is bounded by $|\mathcal{A}_q|V_N(z)(1+\eta_+)(F(s_q) + 106\varepsilon e^2 h(s_q))$. The kernel is nonnegative nondecreasing in q and at most $F_* = 2e^\gamma/k + 106\varepsilon$. Apply (3), using $1/(q-1) \leq (1-z^{-1})^{-1}/q$, and substitute $q = N^\beta$. Since $dq/(q \log q) = d\beta/\beta$, its normalized half-main term is

$$M_1 = \frac{4(1+\eta_+)(1+2/L)(1+v)}{1-e^{-L/8}} (I + 106\varepsilon J_h(t) + 2F_*/L^2), \quad (11)$$

where

$$I = \frac{e^\gamma}{4t} \log \left(6 \frac{3-8\alpha}{3-18\alpha} \right),$$

$$J_h(t) = \int_{1/8}^{t-1/4} \frac{e^{2-8(t-\beta)}}{\beta} d\beta + \log \frac{1/3}{t-1/4}.$$

The split is exactly $s_q = 2$, where the two definitions of h meet. This derives the integral, rather than treating it as an unexplained constant.

Since $q \geq z$ and every factor of d is $< z$, the map $(q, d) \mapsto qd$ is injective. Hence $\sum_q \sum_d |r(qd)| \leq R_A$. Also $\sum_q |r(q)| \leq R_A$ since $y < H$, and $\sum_{d \leq H/q} \mu^2(d)/\varphi(d) \leq 1.1 \log H < .55L$. Here $H/q \geq 10^9$ throughout. The relative sieve remainders are thus at most $(1 + .55L)R_A$. The mass-centering correction adds at most $(1+\eta_+)V_N(z)F_*R_A$ before taking the half. Together with the positive term, their normalized expense is bounded by

$$E_{AP} = \frac{(1.5 + .275L)2.21}{U_*L^3} + 4(1+\eta_+)(1+v)F_*\frac{2.21}{L^4}. \quad (12)$$

This uses the same averaged R_A for every occurrence, never a GRH pointwise bound for the prime moduli.

6.3 The triple-prime term

Take $w_j = z(1+\delta)^j < y$ and replace the product restriction in each p_1 interval by $w_j p_2 p_3 < N$. Drop $(p_1, N) = 1$, but retain $(p_2 p_3, N) = 1$. The enlarged sequences $\bar{\mathcal{B}}_j$ are disjoint: the unique prime factor below y recovers p_1 , and unique sorted factorization of $n = p_2 p_3$ recovers the remaining factors, including $p_2 = p_3$. Thus their integer coefficients are at most one. If some differences are nonpositive, translate each sequence by a large integer multiple of $P_N(y)$. All relevant divisibilities, masses, and remainders are unchanged, so the nonnegative-index sieve still applies.

For $b = \log u/L$ put

$$H_1(u) = \int_y^{\sqrt{N/u}} \frac{dv}{v \log v \log(N/(uv))} = \frac{\log(2-3b)}{L(1-b)}.$$

Counting the last prime by Lemma 2.2 bounds the total mass by $(1+\delta)(1+4/L)N$ times

$$\sum_{z \leq p_1 < y} \frac{1}{p_1} \sum_{y \leq p_2 < \sqrt{(1+\delta)N/p_1}} \frac{1}{p_2 \log(N/(p_1 p_2))}.$$

The inner kernel is nondecreasing and at most $4/L$; its weighted reciprocal-prime error is $8/L^3$. Extending the integral by logarithmic length $\ell/2$ costs $6\ell/L^2$, since its logarithmic integrand is

at most $12/L^2$. The outer kernel H_1 is nonnegative nonincreasing and $H_1(z) < 1/L$, so its error is $2/L^3$. Finally the sum of $1/p_1$ is at most $J + 2/L^2$, with $J = \log(8/3)$. Consequently

$$|\bar{B}| \leq \frac{(1+\delta)(1+4/L)N}{L} [\bar{c} + 2/L^2 + (J + 2/L^2)(6\ell/L + 8/L^2)],$$

where $\bar{c} = \int_{1/8}^{1/3} \log(2-3b)/(b(1-b)) db$. Apply the upper composite sieve at y and use $e^2 h(3t) = 1$. The normalized half-main term is at most

$$M_2 = \frac{(1+\eta_+)(1+v)}{2} (2e^\gamma/t + 318\varepsilon)(1+\delta)(1+4/L) [\bar{c} + 2/L^2 + (J + 2/L^2)(6\ell/L + 8/L^2)]. \quad (13)$$

Each sequence is a rectangle with the endpoints of Lemma 5.1. Because the integer factors are primes at least y , they are coprime to every $d \mid P_N(y)$. Only deleting primes $p_1 \mid d$ changes the principal character's mass from $|\bar{B}_j|/\varphi(d)$. Its per-modulus difference is at most $N\omega(d)/(z\varphi(d))$. With at most $B(L) = 5L/(24\ell) + 1$ boxes and

$$m(L) = 39v_3 + 108x_3^{16}e^{-x_3}/\log x_3 + 26L^5/x_3^{10} + 88L^5(e^{-L/3} + e^{-L/16}) + 106/x_3^6,$$

the normalized half-error is bounded by

$$E_R = \frac{(1+\delta)512m(L)B(L)}{2U_*L}, \quad E_{\text{del}} = \frac{(1+\delta)1.1L^4e^{-L/8}B(L)}{2U_*}. \quad (14)$$

For E_R , divide each term of the rectangle bound by $XY/\log^3 Y$; $\log Y \geq L/8$ gives the factor 512, and $XY \leq (1+\delta)N$. The function $\xi^{16}e^{-\xi}/\log \xi$ decreases for $\xi \geq x_3$, which handles its second term. For E_{del} , the raw expense per box is at most $1.1NL^2/z$; division by $2\Theta_N$ supplies another L^2 . No factor of L is discarded in this normalization.

7 Ledger, certification, and the half-line

Combining (9)–(14) gives

$$\frac{D_{1,2}(N)}{\Theta_N} \geq M_0 - M_1 - M_2 - E_{AP} - E_R - E_{\text{del}} - E_f, \quad E_f = \frac{L^2}{U_*} (2e^{-L/8} + e^{-2L/3} + e^{-L}). \quad (15)$$

This is the single ledger; no auxiliary floating-point budget enters it.

Proposition 7.1 (Strict starting inequalities). *At $L = L_0$ with (1), all product, support, mass, conductor, and sieve-range conditions in the preceding sections hold. The right side of (15) is strictly greater than $9.8468 \cdot 10^{-7} > 9 \cdot 10^{-7}$.*

Proof. The finite algorithms and interval integrals are specified in the appendix. For clarity, coarser decimal endpoints, all verified against their full Arb intervals, enclose the decisive entries as follows:

Entry	Lower endpoint	Upper endpoint
M_0	7.806926624899798362	7.806926624899798363
M_1	6.500392797042414805	6.500392797042414806
M_2	1.306527039983525481	1.306527039983525482
E_{AP}	$3.1474 \cdot 10^{-10}$	$3.1475 \cdot 10^{-10}$
E_R	$5.8028784809024 \cdot 10^{-6}$	$5.8028784809025 \cdot 10^{-6}$
$E_{\text{del}} + E_f$	0	10^{-2700}
Net margin	$9.8468 \cdot 10^{-7}$	$9.8469 \cdot 10^{-7}$

The support slack in (10) exceeds $4.0153 \cdot 10^{-5}$. The three prime-count gates in Lemma 2.2 are bounded by 10^{-1013} , .000058771, and .000528935, respectively. The entire low-conductor absorption gate is below $1.328 \cdot 10^{-938}$. Also $c_4(N)L^2 < 2.200019$, $w > 50991$, $D_{R0} > 10^9$, and all the stated integer cutoffs and strict signs resolve. The medium-prime bounds of Lemma 3.2, ordered-product bound, and $C_\infty > 0.66$ are regenerated from complete integer prime sieves, not obtained by reusing a GRH certificate. The programs check the full intervals and the coarser endpoints above; rounded decimal display alone is not used as an acceptance test. $\square$

7.1 Analytic propagation

We now keep all rational parameters fixed, rather than recomputing α by the search recipe. The left side of (10) increases, with derivative $\alpha - 12.5/L > 0$ at L_0 and thereafter. Thus every distribution support continues to fit H . The product condition of Section 3 already covers every real upper endpoint, independently of N .

The ordinary AP envelope also propagates. Dots in the next calculation mean derivatives with respect to $\log L$. Put $a_1 = (L - 5)/w$ and $b_1 = a_1(w - 5)/(w - 5 \log w)$. For all $L \geq L_0$,

$$1 < a_1 < 1.01, \quad b_1 > 1, \quad w > 50000.$$

For the upper bound on a_1 , use positivity and increasingness of $.01L - 5.05 \log L + 5$. The other inequalities follow from $\log L > 1$ and $\log w > 1$. Apart from fixed positive coefficients, the nonconstant summands in $L^2 p_1$ have forms

$$\begin{aligned} L^2 w^{-6} \log w, \quad L^2 w^{13/2} (w - 5 \log w)^{-10}, \quad L^2 w^{-7/2}, \quad L^2 w^{3/2} e^{-w/12}, \quad L^2 w^{15/2} e^{-w/6}, \\ L^2 w^{-6} e^{-w/2}, \quad L^2 w^5 e^{-w}, \quad L^2 w^{-8}, \quad L^2 w^{-8} e^{-w/6}. \end{aligned}$$

Their logarithmic derivatives are respectively bounded by $2 - 6 + 1.01/\log w$, $2 + 6.5 \cdot 1.01 - 10$, and $2 - 3.5$ for the first three; the others have either a negative power derivative or a negative exponential contribution $(L - 5)/12$ or larger, dominating their displayed fixed powers. They are all negative. The correction $1 + 1/(L^2 w^3) + 1/((1 - 4/w)L)$ decreases. The last summand of $L^2 c_4$ is $.9L^6 e^{w/2-L}/(w^{10} \log L)$, with logarithmic derivative $6 + (L - 5)/2 - L - 10a_1 - 1/\log L < 0$. Only the constant 2.2 remains fixed. This proves $L^2 c_4 < 2.21$ on the whole half-line, not just at its initial point.

The lower main term M_0 is fixed. Every varying factor in M_1, M_2 , namely $1 + 2/L$, $1 + 4/L$, $(1 - e^{-L/8})^{-1}$, L^{-2} and ℓ/L , decreases. Hence both negative main terms decrease. The positive terms of E_{AP} decrease individually. For rectangles put $x_3 = L/8 - \log 2$. The functions $x_3^{17} e^{-3x_3/64}$, $x_3^{16} e^{-x_3}/\log x_3$, x_3^{-6} , and $L^5 e^{-L/3}$, $L^5 e^{-L/16}$ all decrease at and beyond L_0 . The logarithmic derivative of L^5/x_3^{10} is $5 - 10L/(8x_3) < -5$. Also $B(L)/L = 5/(24\ell) + 1/L$ decreases. Thus E_R decreases. The deletion cost is a positive linear combination of $L^5 e^{-L/8}$ and $L^4 e^{-L/8}$, and E_f consists of decreasing exponential-polynomial terms. The tail and prime-mass gates in Section 2 decrease by the same direct derivatives; all their exponents dominate their fixed polynomial degrees already at L_0 . Every side condition persists. Discrete prime sets and the truncated last box need not vary continuously: the common positive envelopes already cover them. The right side of (15) is therefore bounded below by its certified starting value. This proves Theorem 1.1 under its explicitly stated input.

A Finite arithmetic and reproducibility

The proof assigns programs only finite arithmetic tasks. Theorems about zeros, Rosser weights, mean character sums, and the continuum propagation are analytic dependencies; a finite program does not prove them. The reference implementation uses Python with python-flint, Arb/ACB, and 192-bit working precision. It runs from its own directory:

```
python qrh_chen_reproduce.py
python qrh_chen_positivity_search.py
```

The first is the theorem’s fresh arithmetic reproduction; the second records a bounded parameter exploration and is not needed for the half-line proof. All theorem parameters are declared directly as rationals. No old GRH JSON ledger or GRH distribution evaluator is read by the fresh reproduction.

Complete integer Eratosthenes sieves generate all primes for the medium and ordered-product calculations. The medium block algorithm is exactly Lemma 3.2; multiplication of polynomials uses `arb_poly` and discards only coefficients proved irrelevant there. The support maximum uses integer cross multiplication over adjacent primes. The lower bound for C_∞ uses the full odd prime set through T , multiplied by $(T - 1)/T$, the all-integer telescoping tail. The ordered-product algorithm keeps peaks before each prime jump and troughs after it, not sampled interior values; its analytic tail is stated in Section 3.

There are three interval integrations: the delayed $F(s)$ integral, the first piece of J_h , and the integral $\bar{c}$ (the second J_h piece is a logarithm). They use `acb.integral`, retaining its real enclosure. Logarithms in integrands use the integration API’s analytic flag, so branch boundaries are not silently crossed. The path of the F integral lies in $3 \leq x \leq s < 4$, with $x - 2 > 0$; the J_h path lies in $1/8 \leq b \leq t - 1/4 < 1/3$ with $b > 0$; the $\bar{c}$ path lies in $1/8 \leq b \leq 1/3$, with $b, 1 - b, 2 - 3b > 0$. Thus each integrand is holomorphic on a neighborhood of its compact path and has no endpoint singularity. The piecewise sieve function is split at its exact rational/parameter breakpoint before integration. Arb’s outward rounding encloses elementary evaluations and ACB’s validated integration encloses each entire integral, rather than a floating quadrature subsequently decorated with a tolerance.

`qrh_chen_reproduction.json` contains the fresh prime masses, ordered product, two fixed rational endpoints, and source SHA-256 hashes. `qrh_chen_interface_checks.py` independently reconstructs the deleted-prime normalization and checks the contour/rectangle constants. `qrh_chen_positivity_search.json` specifies the explored integer range $50000 \leq L \leq 51200$, five relative widths, and the exact 10^{-9} ceiling recipe for α . It records every branch outcome; the selected positive endpoint is 51046. The preceding recipe endpoint is negative. This establishes neither continuous-parameter nor global optimality. These exploration parameters are frozen, not adapted, in Theorem 1.1.

B Notation and external-input map

Symbol	Meaning
L, L_0	$\log N$ and its fixed starting value 51046.
α, t, D	Support reserve, $t = 1/2 - \alpha$, and large Rosser level N^t .
z, y	Chen cutoffs $N^{1/8}, N^{1/3}$.
$\mathcal{P}_N, P_N(x), V_N(x)$	Odd primes not dividing N , their sieve product, and model density product.
T, w_0, Q_*	Medium cutoff, exact small cutoff, and complete exact odd product 15.
D_0, κ	Defining medium Rosser level and support contraction 11/169.
η_+, η_-	Separate normalized medium upper/lower gap budgets.
ε, F, f, h	Dimension-one tolerance and explicit linear-sieve functions.
x_1, w, H	N/L^5 , its logarithm, and average AP modulus support.
$r(d), R_A$	Centered ordinary AP remainder and its full squarefree sum.
X_3, x_3, D_{R0}, v_3	Fixed rectangle reference point, its logarithm, low-conductor cutoff, and character envelope.

Symbol	Meaning
w_j, X, Y, Z	Small-prime box endpoint and prime–integer rectangle bounds.
U_N, U_*, Θ_N	Singular-series normalization, uniform lower bound, and units of the ledger.
M_0, M_1, M_2	Positive lower main term and the two half-weighted negative main terms.
$E_{AP}, E_R, E_{\text{del}}, E_f$	All ordinary, rectangle, model-deletion, and finite combinatorial charges.

The carrying external inputs are (Z) from [1, Theorem 1.1]; large-prime Rosser masses from [2, Theorem 6, Proposition 13, Table 1]; the unconditional reciprocal-prime tails from [2, Lemmas 16–17] and [4, Lemma 3.2]; squarefree, totient, and divisor-count bounds from [2, Lemmas 22–24]; the high-conductor estimate (7); the multiplicative large-sieve inequality explicitly restated in Section 5; and (9). Every modification of a published sieve range is proved in the text, not effected by dropping a printed hypothesis without a replacement.

References

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